

-G-C bonds are much more resistant to denaturation than A-T rich region

69. The chemical reactions in the small intestine take place in a basic (alkaline) environment. Which number indicates this basic pH?

-8.5

70. Which of the following mutations would produce a minor thalassemia?

-Deletion of one B-globin locus

71. Dietary fats after absorption appear in the circulation as...

-Chylomicron

72. Which of the following cellular components retains the greatest structural integrity throughout the cell cycle?

-Nucleolus

73. Which vitamin is a methyl group acceptor?

-Vitamin B12

74. Which factors are considered in Goldman equation?

-Concentration and permeability

75. Which second messenger activates protein kinase A?

-DAG

76. Which of the following is not a clinical outcome of niacin deficiency?

-Night blindness

77. Which of the following is correctly matched with the vitamin it is derived from?

-Pyridoxal phosphate Vitamin B6

78. Zymogen or proenzyme is a...

-Enzyme precursor

79. A 60-year old female presents with severe back pain for the past week, she is diagnosed with osteoporosis, a condition resulting from calcium depletion in bones. As treatment which of the following vitamins can be prescribed?

-Vit D

4. Deficiency of which of the following vitamins results in megaloblastic anemia?

a. Folate

5. All of the following are functions of vitamin A, EXCEPT?

- a.Maintenance of vision
- b.Maintenance of reproduction
- c.Differentiation and gene expression
- d.Maintenance of calcium balance in plasma

6.A 35-year-old female, a known alcoholic, has been brought to the emergency room. She has been diagnosed with Cardiac Beri-Beri. Which of the following reactions might have been affected?

- c.Pyruvate to lactate

7.Scurvy is due to impaired synthesis of?

- c.Collagen, osteoid, and dentine

8.Which of the following vitamins is an integral part of Coenzyme A (COASH)?

- d.Pantothenic acid

10.Which of the following information is responsible to specify the three-dimensional shape of a protein?

- a.The protein's peptide bond
- b.The protein's amino acid sequence
- c.The protein's interaction with other polypeptides
- d. The protein's interaction with molecular chaperons

11.A point mutation in the beta-globin gene changing the codon from glutamate to valine will likely cause what disease?

- a.Sickle cell anemia

12.When will hemoglobin have the highest affinity for oxygen?

- b.When carbon monoxide levels are high

13. Collagen and elastin have the following similarity:

a. Both are triple helices

14. The pH of the buffer solution depends upon the concentration of?

c. Weak acid

15. The defect in which of the following steps in collagen biosynthesis is responsible for Scurvy?

a. Glycosylation of some residues in collagen

b. Hydroxylation of Lysine

c. The genetic impairment in gene encoding collagen Type II

d. The replacement of Glycine with an amino acid with branched-chain in collagen

16. Which amino acids, can function as an appropriate buffer in our body

d. Histidine

17. Of the 20 standard amino acids, only.... is not optically active. The reason is that its side chain....

b. Glycine; is a hydrogen atom

18. When two carbohydrates are epimers:

d. They differ only in the configuration around one carbon atom.

19. Which of the following monosaccharides is not a carboxylic acid?

b. Gluconate

20. Starch and glycogen are both polymers of:

c. α -D-glucose.

21. Which of the following is a heteropolysaccharide?

d. Hyaluronate

22. In glycoproteins, the carbohydrate moiety is always attached through the amino acid residues:

a. Asparagine, serine, or threonine.

23. Which of these molecules has biological roles including viscosity, lubrication of extracellular secretions?

a. Trehalose.

c. Peptidoglycan.

b. Chitin.

d. Hyaluronate.

24. Absence of which of the following molecules from the lungs of premature infants causes respiratory distress syndrome?

c. Phosphatidylcholine.

25. Which types of Eicosanoids cause bronchoconstriction as well as being potent pro-inflammatory agents, and play a part in asthma?

c. Prostaglandins (PGs).

26. Which of the following statements about membrane lipids is true?

a. Glycerophospholipids are found only in the membranes of plant cells.

b. Glycerophospholipids contain fatty acids linked to glycerol through amide bonds.

c. Some sphingolipids include oligosaccharides in their structure.

d. Triacylglycerols are the principal components of erythrocyte membranes.

27. Which of the following is not true of sterols?

a. Cholesterol is a sterol that is commonly found in mammals.

b. They are commonly found in bacterial membranes.

c. They are precursors of steroid hormones.

d. They have a structure that includes four fused rings.

28. Which of the following about fatty acids is not true?

a. C18 fatty acid with one double bond is called octadecenoic acid

b. Carbon atoms 2 and 3 are often referred to as α and β , respectively.

c. The 16- and 18-carbon fatty acids are most common.

d. Short-chain length and saturation enhance the fluidity of fatty acids.

29. Which of the following statement about lipoproteins is CORRECT?

a. The major apolipoproteins of HDL (α - lipoprotein) are apo A

b. The main apolipoproteins of LDL (β - lipoprotein) are apo B (B-48).

c. Chylomicrons contain apo B-100 only.

d. Apo C and E are integral and cannot be removed.

30. Which one of the following statements correctly describes allosteric enzymes?

a. Effectors may enhance or inhibit substrate binding

b. They are not controlled by feedback inhibition

c. The regulatory site is the catalytic site

d. Michaelis-Menten kinetics describe their activity

31. How do V_m and K_m change in non-competitive inhibition of an enzyme-catalyzed reaction?

d. Decrease V_m , unchanged K_m

32. Which of the following statements about Michaelis-Menten constant is correct?

a. A high K_m indicates a low affinity of an enzyme for its substrate.

33.What is the name of an enzyme catalyzing rearrangement of substrate groupings without altering molecular weight?

- a.Epimerase
- c.Dehydrogenase
- b.ligase
- d.hydrolase
- e.isomerase**

36.A promoter site on DNA

- b.Initiates transcription

37.Given that the chromosomes of mammalian cells maybe 20 times as large as those of Escherichia coli, how can replication of mammalian chromosomes be carried out in just a few minutes?

- a.Eukaryotic DNA polymerases are extraordinarily fast compared with prokaryotic polymerases
- b.The higher temperature of mammalian cells allows for an exponentially higher replication rate**
- c.Hundreds of replication forks work simultaneously on each piece of chromosomal DNA
- d.A great many different RNA polymerases carry out replication simultaneously on chromosomal DNA

38.Patients with hereditary nonpolyposis colon cancer [HNPCC] have genes with microsatellite instability, that is, many regions containing abnormal, small loops of unpaired DNA. This is a result of a mutation affecting ?

- a.Mismatch repair

39.Following ultraviolet damage of DNA in skin,

- d.Endonuclease removes the strand

40.Which of the following descriptions of DNA replication is not common to the synthesis of both leading and lagging strands?

- a.RNA primer is synthesized
- b.DNA polymerase III synthesizes DNA

c. Topoisomerase continuously unwinds duplex DNA at the replication

d. Nucleoside monophosphates are added in a 5' to 3' direction along the growing DNA chain

41. Which of the following enzymes can polymerize deoxyribonucleotides into DNA?

b. DNA ligase

42. Which of the following statements correctly describes the recombinant DNA tool known as plasmids?

d. They sometimes enhance bacterial resistance to antibiotics

43. Which is the most correct sequence of events in gene repair mechanisms in patients without a mutated repair process?

a. Nicking, excision, replacement, sealing, recognition

44. Mammalian chromosomes have specialized structures with highly repetitive DNA at their ends (telomeres). Which aspect of telomeric DNA replication is different from that of other chromosomal regions?

a. The DNA polymerase uses an RNA primer but does not degrade it

b. The DNA polymerase contains an RNA molecule that serves as a template for DNA synthesis

c. The DNA polymerase must cross-link the 5' and 3' termini

d. The DNA polymerase has a σ subunit that facilitates binding to repetitive DNA

45. Radiation therapy is employed for many cancers, including irradiation of the central nervous system to destroy lymphoblasts in leukemia. Which of the following accounts for the destruction of rapidly growing cells?

a. Cross-linking of DNA

46. The DNA sequence M, shown below, is the sense strand from a coding region known to be a mutational "hot spot" for a gene. It encodes amino acids 21 to 25. Given the genetic and amino acid codes CCC = proline (P), GCC = alanine (A), TTC = phenylalanine (F), and TAG = stop codon, which of the following sequences is a frame-shift mutation that causes the termination of the encoded protein? M: 5'-CCC-CCT-AGG-TTC-AGG-3'

b. -CCA-CCC-TAG-GTT-CAG

47.The removal of introns and subsequent self-splicing of adjacent exons occur in some portions of primary ribosomal RNA transcripts. The splicing of introns in messenger RNA precursors is...

c.Carried out by spliceosomes

48.The lactose operon is negatively controlled by the lactose repressor and positively controlled by which of the following?

d.Decreased concentrations of glucose, increased concentration of CAMP

49.Which of the following statements regarding eukaryotic cells is TRUE?

c.Cycloheximide blocks elongation during translation

50.Erythromycin is the antibiotic of choice when treating respiratory tract infections in legionnaire's disease, whooping cough, and Mycoplasma-based pneumonia because of its ability to inhibit protein synthesis in certain bacteria by

a.Inhibiting translocation by binding to 50S ribosomal subunits